

Clinical neuroscience

Detecting neuromagnetic synchrony in the presence of noise

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HIGHLIGHTS

- The signal-to-noise ratio is the most limiting factor for coherence detection.
- Coherence detection may require much larger signal-to-noise ratio than signal detection.
- Coherence detection depends strongly on the underlying model of signal generation.
- Coherence detection is most successful in case of true phase synchronization.

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ABSTRACT

Background: Synchrony between neuroelectric oscillations in distant brain areas is currently used as an indicator of functional connectivity between the involved neural substrates. Coherence measures, which quantify synchrony, are affected by concurrent brain activities, commonly subsumed as noise.

New method: Using Monte-Carlo simulation, we analysed the properties of circular statistics and how those are affected by noise. We considered three different models of neuroelectric signal generation, which are an additive model, phase-reset, and reciprocal phase-interaction. Using the receiver-operating characteristic method, we compared the performances of currently implemented algorithms for coherence detection such as phase-coherence or phase-locking factor, magnitude-squared coherence, and phase-lagging index, all based on circular statistics, and a more general approach to synchrony, using measures of mutual information. We compared inter-trial coherence as a method for signal detection with coherence between multiple sources as measure of source interaction and connectivity.

Results: Charts of performance characteristics showed that the choice of methods depend on the underlying signal generation model. Detection of coherence requires in general a higher signal-to-noise ratio than detection of the signal itself, and again, the difference in performance depends strongly on the underlying model of signal generation.

Comparison with existing methods: Previous comparisons of the performances of different algorithms for signal detection and coherence have not considered systematically the underlying neural generation mechanisms.

Conclusion: Detection of coherence generated by additive signals or a phase-reset requires largely higher signal-to-noise ratio compared to signal detection. Only in case of true phase interaction, signal detection and coherence measures are similarly sensitive.

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1. Introduction

Synchrony between neuroelectric oscillations, recorded with magnetoencephalography (MEG) and electroencephalography

(EEG), has been introduced as an effective mechanism for the communication between distant brain areas (Fries, 2009; Singer, 1999; Uhlhaas et al., 2009). In studies of normal and disordered brain function, measures of synchrony have provided rapidly growing evidence for a close relation between oscillations at beta and gamma frequencies and a variety of cognitive functions such as perceptual grouping (Engel and Singer, 2001), focused attention (Jensen et al., 2007), maintenance of information in short-term memory (Howard et al., 2003), multi-sensory integration (Hermer-Vazquez et al., 2007), formation of associative memories (Headley and Weinberger, 2011) and sensory motor coordination

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(Womelsdorf and Fries, 2006). Gamma synchrony has been shown to characterize functional and behavioral changes in these different cognitive functions (Herrmann et al., 2010). Recently, cross-frequency coupling, in which bursts of high-frequency activity occur preferentially during a specific phase of slower oscillations, has been shown as a mechanism that controls cortical computation such as information encoding and stimulus selection (Belluscio et al., 2012; Canolty and Knight, 2010; Cohen et al., 2009; Jensen and Colgin, 2007; Sauseng et al., 2008).

However, measures of synchrony within or between frequencies are challenged by the presence of concurrent brain activity, which is not related to the focus of interest and often subsumed as “noise”. Given the significance of synchrony measures for functional brain connectivity, strikingly little is known about how noise affects the sensitivity of detecting synchrony. Most common approach for separating signal and noise is repeating the experimental trial multiple times and using statistical methods for identifying effects in mean. Signal averaging is the most prominent example for such statistics in the time domain analysis (König et al., 2015). Because the signal phase plays an important role for synchrony measures, circular statistics have been proven to be a powerful tool for signal detection in the frequency domain (Lachaux et al., 1999). The sensitivity of phase statistics related to noise and number of averages has been studied specifically for phase-locking to a stimulus event across repeated trials (Dobie and Wilson, 1993). Such inter-trial phase-coherence is a special case of synchrony between the brain signal and the known phase of an external stimulus. In our study, we expand the performance analysis of inter-trial phase coherence to the general case of synchrony between multiple brain signals. This approach was inspired by the fact that often we have a priori information about the magnitudes of signal and noise and thus a reasonable good estimate for the sensitivity of signal detection. Now we are asking whether detectability itself is sufficient for demonstrating synchrony or whether phase synchrony might even be a more sensitive tool.

Measures of phase coherence have been directly linked to the signal-to-noise ratio (SNR) (Dobie and Wilson, 1989). We aim at continuing this approach by asking how much SNR is required for successful detection of synchrony. For such performance analysis we have to consider the neural mechanism underlying synchrony and the methods of measuring synchrony. Therefore, we will address how currently discussed models of generating synchronous oscillations affect synchrony measures. Moreover, synchrony between two neural populations may result from interaction between those populations or alternatively from synchronization with a common input; and we will consider the effects of different network topologies. Also, some algorithms can detect synchrony better than others; and we are comparing the performance of multiple measures of synchrony using Monte-Carlo simulations.

1.1. Signal models

Modeling the electromagnetic brain signal is inextricably connected to a long-standing controversial discussion about the generation mechanism of the event-related signal changes in EEG and MEG. Initially it has been assumed that an external stimulus or internal cognitive event elicits neural activity in addition to ongoing brain signals. This additive model has been challenged by the alternative view that the stimulus event causes merely a reorganization of the phases of ongoing activities (Fell et al., 2004; Hanslmayr et al., 2007; Makeig et al., 2002; Sayers et al., 1974). Subsequent studies aimed to distinct between the additive and phase-reset models based on whether event-related responses were accompanied by an increase in signal power, thus supporting the additive model (Shah et al., 2004), or the responses occurred without a change

in signal power and thus providing evidence for the phase-reset model (Mormann et al., 2005). However, it has been cautioned that such discussion may not be conclusive and both generation mechanisms should be considered (Sauseng et al., 2007; Shah et al., 2004). Even further mechanisms were proposed for the generation of slow wave responses based on baseline shift (Nikulin et al., 2007) and asymmetry (Mazaheri and Jensen, 2008) of alpha oscillations. Modeling studies demonstrated neural synchrony at microscopic level as the origin of evoked responses and showed consistent macroscopic signals when considering either the additive or phase-reset mechanism (Telenczuk et al., 2010). Both mechanisms may exist in parallel, however may have different impact on synchrony measures. Therefore, we will incorporate the additive and phase-reset model in our simulation study. A third, likely most important, mechanism is that two brain areas interact with each other in a way that the phase in one area changes its time course to follow the phase in the second area. Such ‘true synchronization’ by reciprocal interaction between sources (Burgess, 2013) does not result in consistent phase locking to the stimulus event and therefore does not elicit an evoked response neither an event related power change. The relations between the three signal generation models and event-related changes in signal power, evoked response, and synchrony are depicted with the schematics in Fig. 1. This figure demonstrates a scheme of the three models: The additive model is characterized by signal power increase and stimulus related phase reorganization, the phase-reset model shows phase reorganization only, and the phase interaction does not show any effects, which could be seen in the stimulus locked averaged signal. Common to the three models is that they generate synchrony, measured as coherence.

A prerequisite for the performance analysis is the definition of synchrony. The strict definition of synchrony seems to be fulfilled by the model of reciprocal interaction between distinct neural substrates that generate autonomous oscillations and adjust their rhythms through weak coupling (Rosenblum and Pikovsky, 2003). However, the underlying generation model is commonly not known. Therefore, we accept synchrony more generally based on the observed signals if the phases of two signals evolve in parallel over a certain time interval

$$|\theta_1(t) - \theta_2(t)| < \text{const}, \quad t \in [t_1 \dots t_2] \quad (1)$$

where $\theta_{1,2}(t)$ are the phases of the two signals $x_1(t)$ and $x_2(t)$ (Mormann et al., 2000). This definition applies genuinely to phase synchronization but also to a phase reset, which forces the two signals to oscillate with same phase; as well it applies to superimposition of a common signal thus making same phase more likely in both signals. The definition of synchrony can be expanded to two oscillations at different frequencies with ratios n/m ,

$$|n\theta_1(t) - m\theta_2(t)| < \text{const}, \quad (2)$$

in which n and m are small integers and $\theta_{1,2}$ are the phases of the two signals (Tass et al., 1998). Functionally relevant synchrony may exist over time intervals as short as 100 ms only and then changes considerably when switching between micro states (Lehmann et al., 1987). In addition to phase synchrony within trials the consistency of phase values across repeated trials is used as a measure of synchrony and has been first introduced as ‘phase-locking factor’ (PLF) (Tallon-Baudry et al., 1997) but is also synonymously termed ‘inter-trial coherence’ (ITC) (Makeig et al., 2004). This measure of synchrony can be applied to test whether the phase difference between two signals is consistent across trials. Although the measures are very similar, a different term ‘phase-locking value’ (PLV) had been introduced for the latter (Lachaux et al., 1999). The ITC has been used since the 1980’s as a statistical method for detecting an event-related signal (Jervis et al., 1983), whereas the PLV identifies synchrony between neural sources.

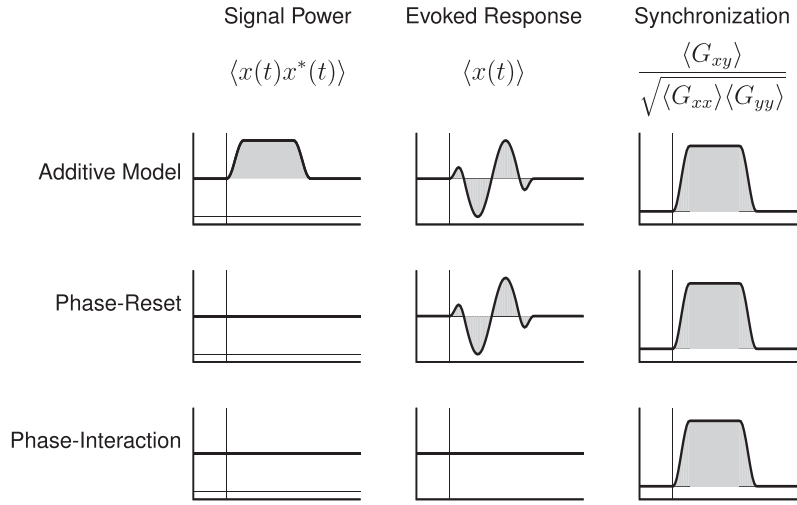


Fig. 1. Schematics of the relations between the additive, phase-reset, and reciprocal phase interaction models and event-related changes in signal power, evoked response and synchronization.

Commonly, we assume band-limited processes for phase analysis because the signal phase is meaningfully defined for this case only. Moreover, brain activities in narrow frequency bands have unique meanings and are discussed separately. The signal of interest $x(t)$ can be obtained from bandpass filtering or time-frequency representations like the wavelet transform. Communality and differences between the time-frequency analyses have been investigated previously (Bruns, 2004). Generally, we describe the signal in the complex domain as

$$x(t) = A(t) \cdot e^{i(\omega t + \varphi(t))}, \quad (3)$$

in which $A(t)$ is the time varying amplitude, ω the center frequency of the bandpass or the wavelet and $\varphi(t)$ the phase deviation from otherwise monotonously increasing phase ωt . Omitting the steady oscillation at ω results in the more convenient phasor notation

$$X(t) = A \cdot e^{i\varphi(t)}, \quad e^{i\varphi(t)} = \cos \varphi(t) + i \cdot \sin \varphi(t) = \frac{r(t)}{|r(t)|}. \quad (4)$$

The random phase deviation $\varphi(t)$ can be modeled as a complex jointly-Gaussian random vector $r(t)$ with same normal distributions for the real and imaginary parts. Given such signal description, the additive model for the EEG or MEG signal can be expressed as

$$Y(t) = S + N \cdot e^{i\varphi(t)}. \quad (5)$$

The ratio between the signal amplitude S and the standard deviation of the noise N defines the SNR. For synchrony between two signals $x_1(t)$ and $x_2(t)$ caused by an additive mechanism, we assume the same signal S superimposed to independent noise contributions to both signals

$$Y_i(t) = S + X_i \cdot e^{i\varphi_i(t)}, \quad i = [1, 2]. \quad (6)$$

In this case, the superimposed signal S could result from direct interaction between the sources x_1 and x_2 or from common interaction with a third source.

The alternative model of phase reset assumes that the phase is entirely random before a stimulus event, it is set to a constant value at time $t=0$, and then it decays over time into a random process again (Sauseng et al., 2007). Concurrent noise can be considered by a partial phase reset, and we can describe the event-related signal change as manipulation of the phase,

$$Y(t) = \text{abs}(X(t)) \cdot \frac{r(t)}{|r(t)|}, \quad r(t) = a \cdot e^{i\theta} + (1-a)e^{i\varphi(t)}. \quad (7)$$

The parameter a indicates the partial of ongoing oscillations that is reset to the constant phase θ and thus is closely related to the SNR. The parameter a ranges between 0 and 1. Zero means no reset at all, and $a=1$ relates to a complete reset or complete locking to the stimulus phase.

For the third model, we assume that synchrony results from reciprocal phase interaction: The phase of a signal $x_1(t)$ is partially replaced by the phase of $x_2(t)$ and vice-versa,

$$Y_1(t) = \text{abs}(X_1(t)) \cdot \frac{r_1(t)}{|r_1(t)|}, \quad r_1(t) = a \cdot e^{i\varphi_2(t)} + (1-a)e^{i\varphi_1(t)}. \quad (8)$$

Again, the parameter a in the interval between 0 and 1 determines the degree of partial synchronization and could be termed phase-mixing factor. For reciprocal interaction, the phase coupling is maximal for $a=0.5$. For unilateral interaction, in which x_1 keeps its phase and x_2 morphs partially into the phase of x_1 , the coupling is strongest for $a=1$. More realistic in noisy MEG and EEG data, values of a are much smaller than 0.5.

1.2. Algorithms for measuring synchrony

Most common methods for calculating synchrony are using circular statistics (Fisher, 1995; Mardia, 1972), namely the circular mean. The ITC is calculated as the sample mean of the complex amplitude $X(t)$, normalized for each trial.

$$\text{ITC}_{\text{PC}} = \left| \left\langle \frac{X(t)}{|X(t)|} \right\rangle \right|, \quad (9)$$

in which $\langle \cdot \rangle$ denotes the sample mean and $|\cdot|$ the absolute value. This type of ITC is generically termed phase coherence (PC) because phase information only is used. An equivalent however more explicit equation for the ITC is

$$\text{ITC}_{\text{PC}} = \sqrt{\left(\frac{1}{n} \sum_{i=1}^N \cos \varphi_i \right)^2 + \left(\frac{1}{n} \sum_{i=1}^N \sin \varphi_i \right)^2}. \quad (10)$$

This is the common description of the test statistics for the Rayleigh test (Fisher, 1995). Alternatively, the amplitude

variation across trials is preserved in the magnitude-squared coherence (MSC)

$$ITC_{MSC} = \frac{|\langle X(t) \rangle|}{\langle |X(t)| \rangle} \quad (11)$$

Again in more explicit notation (Dobie and Wilson, 1989)

$$ITC_{MSC} = \frac{\left(\left((1/n) \sum A_i \cos \varphi_i \right)^2 + \left((1/n) \sum A_i \sin \varphi_i \right)^2 \right)}{(1/n) \sum A_i^2}. \quad (12)$$

Coherence between two signals X and Y can be measured using the cross spectral density

$$G_{xy} = \langle X \cdot Y^* \rangle. \quad (13)$$

in which the asterisk denotes the conjugate complex. An elementary transform shows that the cross spectral density is essentially an estimate of the mean phase difference

$$G_{xy} = \langle A_1 e^{i\theta_1} \cdot A_2 e^{-i\theta_2} \rangle = \langle A_1 A_2 e^{i(\theta_1 - \theta_2)} \rangle. \quad (14)$$

For obtaining coherence measures, the cross-spectral density is related to an estimate of the signal power or the auto density

$$G_{xx} = \langle A_i A_i^* \rangle. \quad (15)$$

Depending on how the cross spectral density is normalized, two types of coherence measures are widely used. First, the amplitude information can be removed for each trial, obtaining a phase-coherence type of the phase-locking value PLV

$$PLV_{PC} = \left\langle \frac{G_{xy}}{|G_{xy}|} \right\rangle. \quad (16)$$

Alternatively, estimates of the cross-spectral density and the auto spectra can be calculated separately across trials, resulting in a magnitude-squared type of the PLV

$$PLV_{MSC} = \frac{\langle G_{xy} \rangle}{\sqrt{\langle G_{xx} \rangle \langle G_{yy} \rangle}}. \quad (17)$$

The coherence measure PLV is generally a complex number. A significant real part and vanishing imaginary part indicates zero mean phase difference and could be interpreted as synchrony without time delay. Then again, non-zero imaginary coherence indicates a time delay between X and Y . This property can be useful for differentiating between coherence resulting from physical cross-talk, which occurs without noticeable time delay, and real neural communication between sources, involving some conduction delay (Nolte et al., 2004). Therefore some authors use the term ‘coherency’ for the complex valued measure and ‘coherence’ for its absolute value (Sander et al., 2010).

It has been argued that solely considering the imaginary part of coherency may be insensitive in case of a large real part resulting from physical cross-talk because the real part contributes to the signal power and thus the denominator when normalizing the cross-spectral density. Therefore, a sign test on the phase difference, equivalent to a sign test on the imaginary part of cross-spectral density has been proposed as more sensitive alternative and has been termed phase-lag index (PLI) (Stam et al., 2007)

$$PLI = |\langle \text{sign}(\Delta\varphi) \rangle| \quad (18)$$

or equivalently

$$PLI = |\langle \text{sign}(\text{Im}(G_{xy})) \rangle|. \quad (19)$$

The PLI measure has been further developed by considering the actual amount of phase difference by introducing a weighting factor, which is largest for phase differences of $\pm\pi/2$ and smallest for

phase differences of zero and π (Vinck et al., 2011). The effects of noise, cross-talk and the amount of phase difference on the performance of detecting synchrony are widely unknown. Therefore we included the PLI method into the performance analysis and comparison between common algorithms for coherence measures.

So far, all introduced measures are based on circular statistics. More general concepts of synchrony have been developed, which may specifically consider non-linear signal dynamics. A method of detecting synchrony in noise based on mutual information has been introduced by (Paluš, 1997). Presenting the phases of two signals $x(t)$ and $y(t)$ on the x -axis and y -axis results in a two-dimensional phase-space representation of the signals. If both phases are independent, the phase space will be filled uniformly. On the contrary, if both signals contain some dependency some regions of the phase space will be more frequently covered than others. In the extreme case of equal phases, the phase-space representation collapses to the main diagonal. Mutual information measures can be used to test whether the signal phases cover the phase space differently from a uniform distribution. Let $p_1(\theta_1)$ and $p_2(\theta_2)$ be the probabilities of the phases θ_1 and θ_2 respectively, and $p_{1,2}(\theta_1, \theta_2)$ their joint probability, the mutual information I is given as

$$I(\theta_1, \theta_2) = \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} p_{1,2}(\theta_1, \theta_2) \log \frac{p_{1,2}(\theta_1, \theta_2)}{p_1(\theta_1) p_2(\theta_2)} d\theta_1 d\theta_2. \quad (20)$$

The mutual information I is zero if both phases are completely independent and is larger than zero in case of synchrony. The described concept had been extended from the two-dimensional signal representation to a multi-dimensional state-space representation using time-delay embedding (Arnhold et al., 1999) for the measure of generalized synchronization likelihood (Stam and Van Dijk, 2002).

2. Material and methods

2.1. H_0 -distribution of phase statistics

The circular mean is a non-consistent statistics, it decreases with increasing sample size (Maris et al., 2007), which complicates the direct comparison between reported phase statistics. Therefore, we established first empirical estimates of the H_0 -distribution and critical values for phase coherence for various sample sizes. Complex random-sample vectors were generated from independent normal-distributed random processes for the real and imaginary parts using the Matlab function **randr**. The complex vectors were projected onto the unit circle and the circular mean was calculated for $n = 25$ to $n = 800$ trials with a total of 10^6 samples for each n . Probability density graphs for the circular mean $\bar{R}$ and the derived statistics $n\bar{R}^2$ were constructed from histograms of the permutations. The same procedure was applied to the PLV measure. One purpose of estimating the H_0 -distribution is obtaining critical values. A numerical method for critical values of the circular mean has been developed by Greenwood and Durand (1955). It has been shown that a simplified approximation is valid for $n > 50$, which is the case in most EEG and MEG studies and obtains p -values from the phase statistics as $p = \exp(-n\bar{R}^2)$ (Fisher, 1995, p 70). For using this approximation, the H_0 -distributions for the circular mean $\bar{R}$ were transformed into distributions of the derived statistics $n\bar{R}^2$.

2.2. Phase statistics under H_1

Samples of the phase statistics under the alternative hypothesis H_1 were constructed according to the three data models, described with equations (6, 7, and 8). The parameter SNR for the additive model was chosen with 50 logarithmically spaced

values between 0.01 and 10. Similarly, the phase-mixing parameter a for the phase-reset and the phase-interaction models was chosen with 50 logarithmically spaced values between 0.01 and 1.0. Phase statistics were calculated for $n = (25, 50, 100, 200, 400, 800)$ using magnitude-squared coherence (11). A total of 10^6 samples were generated for each data model, number of trials and choice of the SNR or a parameter, respectively. For the additive model and the phase-reset model, both ITC and coherence measures were obtained from sets of single samples or pairs of samples, respectively. Characteristics of the $n\bar{R}^2$ statistics as function of the SNR were created. For the phase-reset model, we calculated SNR values from the samples for the ITC statistics as the ratio of the sample mean and the standard deviation, which translated the scale of phase-mixing factor a into a SNR scale. Characteristics of the $n\bar{R}^2$ statistics for the phase-reset model were created as function of the estimated SNR. Accordingly, the results for the phase-interaction model were shown on the same SNR scale.

2.3. Experimental example for the relation between SNR and ITC

Phase coherence and SNR were calculated for the auditory evoked transient gamma band response in a retrospective analysis of a previously reported study (Ross and Tremblay, 2009). The auditory gamma band response is a small oscillatory response, centered around 40 Hz, which was elicited by the onset of the sound of a syllable 'ba' in 42 participants in the age range between 19 and 74 years. MEG data were recorded repeatedly in multiple sessions, thus for each participant 2400 stimulus related epochs of bilateral auditory cortex source activity were available for analysis. The data in the latency interval of 0 to 100 ms were bandpass filtered between 28 Hz and 48 Hz, submitted to a Hilbert transform, and averaged across all trials for each participant. Individual peak latencies were determined and ITC_{PC}, signal amplitude and the standard deviation were estimated from single trial analysis at the peak latency. The SNR was calculated as the ratio of the signal amplitude and the standard deviation. Mean, quartiles, and 2.5 and 97.5 percentiles of the distribution of the SNR were calculated.

2.4. Receiver-operation characteristics analysis

Performance analysis was conducted according to the Monte-Carlo method of repeated random sampling of various coherence measures under the null hypothesis H_0 and the alternative H_1 according to the different data models and obtaining the empirical probability density functions (PDF). The PDF _{H_0} , obtained under H_0 , and PDF _{H_1} overlap more or less strongly. If a decision is made at the level x_0 , the area under PDF _{H_1} for all $x > x_0$ represents the hit rate and the area under PDF _{H_0} for all $x > x_0$ represents the false alarm rate. Plotting the hit rate, the probability of correct detection, versus the false alarm rate for all levels of x_0 constructs the receiver-operating characteristics (ROC) (Fawcett, 2006). Empirical cumulative probability density functions (CDF) were obtained from sorting and normalizing the outcome measures of the random samples. The CDFs under H_0 and H_1 were interpolated and resampled on same equidistant samples to construct the ROC. The area under the curve (AUC), which served as a measure of the total probability of detection, independent of the chosen threshold, was calculated by numerical trapezoidal integration using the Matlab function **trapz**. ROCs and the corresponding AUC were calculated for a wide range of SNR values for studying how the SNR affected the detection performance and characteristics of the AUC as function of the SNR were obtained as main outcome of the analysis.

2.5. Performance analysis

Surrogate data were constructed from bandpass filtering random samples using the Matlab function **randn**. The sampling rate was 625 s^{-1} , which is common to MEG recordings. The bandpass filter was an FIR filter constructed with the Matlab function **fir1** with length of 256 samples and $f_1 = 56 \text{ Hz}$ and $f_2 = 112 \text{ Hz}$, thus simulating data in the upper gamma frequency band. A Hilbert transform was applied to the impulse function of the FIR filter. Thus, convolution of the random samples with the complex filter function resulted in complex bandpass filtered time series. Time series were constructed with five different durations in the range of 100 ms to 1600 ms. For each time series, the random sequences were 512 samples longer and the first and last 256 samples were discarded after bandpass filtering to avoid edge effects. A reference sine wave at 79 Hz, the center of the bandpass filter was used as the 'signal' x_0 .

Pairs of simulated signals $y_1(t)$ and $y_2(t)$ were constructed from two bandpass filtered random signals $x_1(t)$ and $x_2(t)$ and the reference sine wave $x_0(t)$ according to the signal models described with Eqs. (6)–(8). For the additive model and the phase-reset model, two different versions of the signal $y_2(t)$ were constructed for calculating ITC and PLV, respectively. With the parameter SNR this yielded for the additive model:

$$y_1(t) = \text{SNR} \times x_0(t) + x_1(t)$$

$$\text{ITC} : y_2(t) = x_0(t) \quad (21)$$

$$\text{PLV} : y_2(t) = \text{SNR} \times x_0(t) + x_2(t)$$

The parameter SNR was varied between 0.001 and 0.5 in 80 logarithmic steps. For the phase-reset model the phase of $y(t)$ was manipulated as combination of the phase $\varphi(t)$ of the surrogate $x(t)$ and the constant phase φ_0 of the reference signal using the phase-mixing parameter a :

$$y_1(t) = |x_1(t)| \cdot \frac{r_1(t)}{|r_1(t)|}, \quad r_1(t) = a \cdot e^{i\varphi_0} + (1-a)e^{i\varphi_1(t)}$$

$$\text{ITC} : y_2(t) = x_0(t) \quad (22)$$

$$\text{PLV} : y_2(t) = |x_2(t)| \cdot \frac{r_2(t)}{|r_2(t)|}, \quad r_2(t) = a \cdot e^{i\varphi_0} + (1-a)e^{i\varphi_2(t)}$$

The parameter a was varied between 0.001 and 0.5 in 80 logarithmic steps covering a similar SNR range as for the additive model. Finally, reciprocal interaction between signal phases was modeled by combining the phases of the two surrogate signals x_1 and x_2 , again using the phase-mixing parameter a in the range between 0.001 and 0.5:

$$y_1(t) = |x_1(t)| \cdot \frac{r_1(t)}{|r_1(t)|}, \quad r_1(t) = a \cdot e^{i\varphi_2(t)} + (1-a)e^{i\varphi_1(t)} \quad (23)$$

$$y_2(t) = |x_2(t)| \cdot \frac{r_2(t)}{|r_2(t)|}, \quad r_2(t) = a \cdot e^{i\varphi_1(t)} + (1-a)e^{i\varphi_2(t)}$$

Sets of signals were generated with trial size of 12, 25, 50, 100, 200, 400, and 800, thus covering typical sample sizes in EEG and MEG studies. ITC and PLV were calculated using phase-coherence (Eq. (16)), magnitude-squared coherence (Eq. (17)), and the phase-entropy measure (Eq. (20)). For the phase entropy, we divided the 2π -range of the phase φ_1 and φ_2 into 20 equal bins and obtained the joint probabilities from counting occurrences of samples in each bin. For studying the performance of the phase-lagging index (PLI), we introduced constant phase differences of $\pi/2$, $\pi/4$, $\pi/8$, $\pi/16$, and $\pi/32$ between y_1 and y_2 . The simulations were performed with repetitions of sets of 200 trials of 200-ms duration. The PLI had been introduced for detecting a significant imaginary part of coherence,

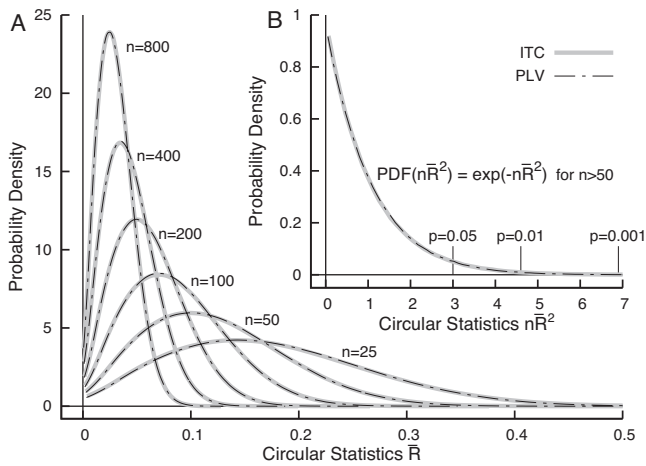


Fig. 2. Distributions of circular statistics under the null hypothesis H_0 . (A) Empirical estimation of the distributions of the circular mean $\tilde{R}$ for various sample sizes n . The dashed lines for PLV overlay almost perfectly the gray lines for ITC, indicating identical distributions for both measures. The mode of the distributions moves toward zero with increasing number of samples. (B) The probability density for the statistics $n\tilde{R}^2$ approximates to $\exp(-n\tilde{R}^2)$ and provides critical values independently of the number of trials.

indicating a time lag caused by neural delay, in the presence of contamination by spatial signal spread affecting the real part only. We modeled such signal spread by introducing parametrically a correlation between the signals y_1 and y_2 in steps of 0.05 between zero and 0.35.

The signal randomization was repeated 20,000 times for each condition and under the null hypothesis, $\text{SNR} = 0$ or $\alpha = 0$, and under H_1 for the various values of the parameters SNR and α . ROC were constructed from the resulting coherence measures and AUC values were the final outcome measure of the simulations.

3. Results

3.1. Distribution of phase statistics under the null hypothesis H_0

The empirical H_0 -distributions for phase statistics, shown in Fig. 2A, reveal that the mode of H_0 moves toward smaller values of $\tilde{R}$ for larger sample size. The characteristics demonstrate the non-consistent behaviour of the phase statistics, and make clear that in general the coherence measures become smaller when the sample size increases. Consequently, critical values depend strongly on the sample size. A further notable feature is that ITC and PLV measures share the same H_0 -distribution although ITC was derived from the signal samples whereas PLV was based on signal power like measures of cross-spectral density. The empirical H_0 -distributions from Fig. 2A had been transformed into distributions of $n\tilde{R}^2$ values, shown in Fig. 2B. The multiple graphs in Fig. 1A for different sample sizes collapse into a single graph, and the probability density function of $n\tilde{R}^2$ is independent of the sample size. From this graph, p -values can be immediately obtained for both ITC and PLV measures.

3.2. Phase statistics under H_1 for the three data models

Empirical estimates of phase statistics are shown for ITC and PLV in as function of the SNR in Fig. 3. The sample size n was parametrically varied between 25 and 800. Phase statistics had been calculated according to the MSC algorithms. For the phase-reset model, the SNR had been obtained as the ratio of the empirical sample mean, corresponding to the evoked response, and the standard deviation of the simulated data. The nominal scale of the

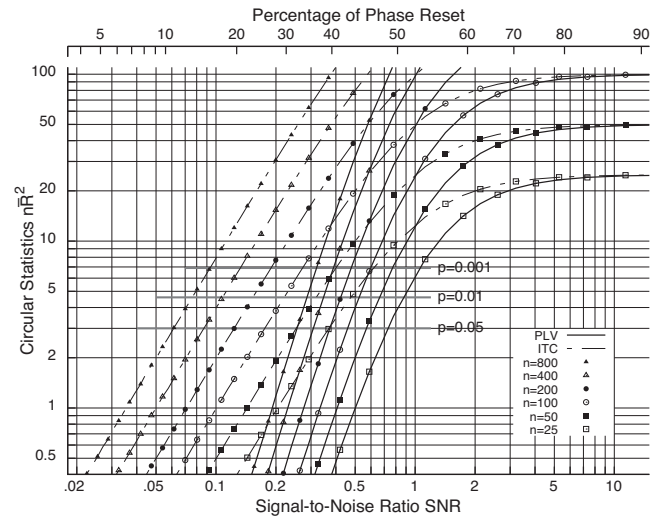


Fig. 3. Estimates of phase statistics for the additive and the phase-reset signal models with parametric variations of sample size n and SNR . The dashed lines for ITC and solid lines for PLV represent approximations to $n\tilde{R}^2$ as function of the SNR , while the symbols represent empirical estimates of the statistics at discrete values of SNR . For the phase-reset model, the scale of percentage of phase reset corresponding to the SNR is shown on top.

percentage of phase reset is given on top of the characteristics in Fig. 3. The $n\tilde{R}^2$ phase statistics were identical for the additive and phase-reset models when using the common SNR scale. Therefore a single figure has been produced for the additive and phase-reset data model.

Although the H_0 distributions were identical for ITC and PLV (Fig. 2), notable differences were found between H_1 characteristics of both measures, indicating substantial performance differences between both. When using ITC measures for signal detection of evoked responses, the dependency of MSC on the SNR has been shown as $\tilde{R}^2 = \text{SNR}^2 / (1 + \text{SNR}^2)$ (Dobie and Wilson, 1989). Dashed lines of the function $n \cdot \text{SNR}^2 / (1 + \text{SNR}^2)$ have been superimposed in Fig. 3 to the symbols, representing the empirically estimated statistics at discrete SNR values. The overlay demonstrates almost perfect matching between empirical values and the approximation. Horizontal gray lines in Fig. 3 indicate critical values at $\alpha = 0.05, 0.01$, and 0.001 and allow direct access to the sample size required at a given SNR or vice versa the required SNR for a given sample size. The $n\tilde{R}^2$ statistics for the ITC approximates the constant value of n for large SNR values and asymptotically $n\text{SNR}^2$ for small SNR values. The latter illustrates that the $\sqrt{n}$ -law of signal averaging applies to the phase statistics: A signal with $\sqrt{k}$ smaller SNR requires k -times larger sample size for detection with equal performance.

For exploring the fundamentally different relation between PLV and SNR , we extended the approximation by Dobie and Wilson (1989) to the magnitude-squared type of the PLV measure, which resulted in $n\tilde{R}^2 = n(\text{SNR}^2 / (1 + \text{SNR}^2))^2$. This function is overlaid by solid lines to the empirically obtained estimates of the PLV values and again shows almost perfect match. For large values of SNR , the PLV graphs approximate against n as it has been observed for ITC. However the different behaviour of PLV compared to ITC becomes obvious for more realistic smaller values of SNR . The $n\tilde{R}^2$ statistics for the PLV approximates asymptotically $n\text{SNR}^4$ for small SNR values. Consequently, when the SNR decreases by a factor $\sqrt{k}$, the same significance level can be obtained only with k^2 larger sample size. Specifically, the graphs in Fig. 3 demonstrate that signal detection using ITC can be successful at SNR in the range of 0.1–0.2 with reasonable sample sizes. However an unpermitted large number

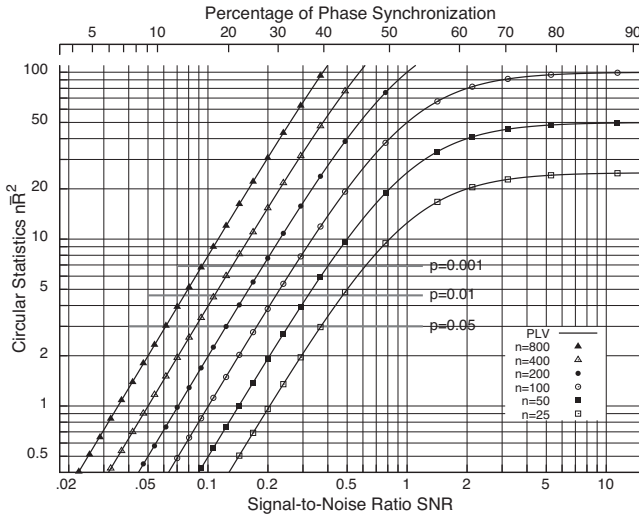


Fig. 4. Estimates of phase statistics for the phase-synchronization model with parametric variations of sample size and the percentage of phase synchronization. For convenient comparison with Fig. 3, the scale of partial synchronization on top of the graph was translated into SNR measures using the same empirical relation between SNR and phase-reset as in Fig. 3. The symbols indicate empirically estimated circular statistics using the MSC version of coherence. The overlaid solid line is the approximation $n\bar{R}^2 = n \cdot \text{SNR}^2 / (1 + \text{SNR}^2)$.

of samples would be required for detection of coherence between two signals at same SNR.

3.3. Phase statistics for the phase-interaction model

Empirical estimates of MSC type of coherence measures are shown for the underlying phase-interaction model in Fig. 4. For convenient comparison, the scale of partial phase synchronization was translated into a SNR scale using the same relation as in Fig. 3. The empirical estimates could be approximated by $n\bar{R}^2 = n \cdot \text{SNR}^2 / (1 + \text{SNR}^2)$, which is the same approximation, which was valid for ITC in case of the additive or phase-reset model. Thus, PLV measures appear to be far more sensitive at small values of partial synchronization under the model of reciprocal interaction than under the assumption of an additive or phase reset model. The fundamental difference can be explained when looking into calculation of the cross-spectral density under the different models. For the additive and phase-reset signal generation mechanism, we modeled the two signals of interest Y_1 and Y_2 as combinations of two random processes X_1 and X_2 with the signal X_0 , using the parameter a ,

$$Y_1 = aX_0 + (1 - a)X_1 \quad \text{and} \quad Y_2 = aX_0 + (1 - a)X_2. \quad (24)$$

The stimulus event, eliciting the phase-reset or the additive component, occurs independently of the phases of both independent signals X_1 and X_2 . This means that all estimation values of combined products vanish:

$$\langle X_1 X_0 \rangle = 0, \quad \langle X_2 X_0 \rangle = 0, \quad \text{and} \quad \langle X_1 X_2 \rangle = 0. \quad (25)$$

When calculating the cross-spectral density $G_{Y_1 Y_2} = Y_1 Y_2^*$, only the term $a^2 X_0^2$ contributes. Thus the cross-spectral density is proportional to a^2 and declines rapidly for small values of a .

Although the formal descriptions of Y_1 and Y_2 look very similar for the case of phase synchrony resulting from reciprocal interactions,

$$Y_1 = aX_2 + (1 - a)X_1 \quad \text{and} \quad Y_2 = aX_1 + (1 - a)X_2. \quad (26)$$

Now, the partial interaction terms aX_2 and aX_1 are no longer independent of X_1 and X_2 . Consequently, the non-vanishing terms

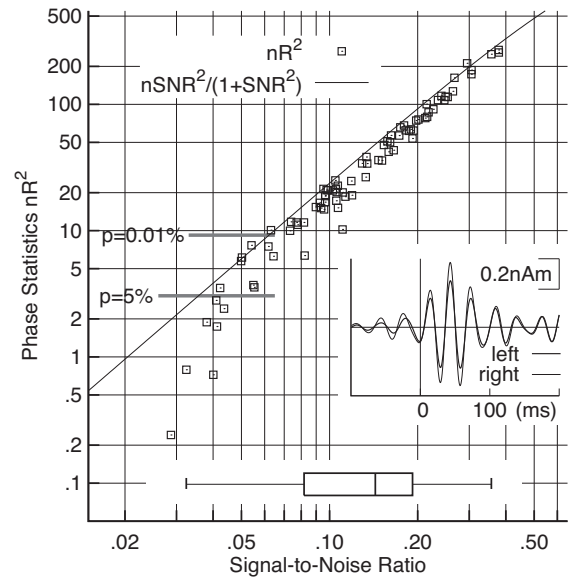


Fig. 5. Experimental example for the relation between ITC_{PC} and SNR. The square symbols indicate individual measures of phase statistics and SNR obtained from a wavelet analysis of auditory gamma band responses. The solid line represents an approximation of the relation between phase statistics and SNR. The insets show waveforms of the grand averaged transient gamma band response for the left and right auditory cortices and a box plot describing the range of SNR measures. The box plot indicates the second and third quartiles, and the median of SNR, with the whiskers showing the 2.5% to 97.5% range.

in the cross-spectral density are $a(1 - a)\langle X_1^2 \rangle$ and $a(1 - a)\langle X_2^2 \rangle$. Thus, the cross-spectral density is proportional to $(a - a^2)$ or roughly proportional to a when neglecting the smaller contribution of a^2 for small values of a .

3.4. Example of the relation between SNR and phase statistics

Before using Monte-Carlo simulation for performance analysis of different coherence measures, we studied the relation between phase statistics and the SNR in a retrospective analysis of MEG data from an auditory evoked response study. The goal was obtaining a realistic range of SNR values for the widely interesting case of gamma oscillations.

Auditory evoked gamma responses are typically elicited with click stimuli or brief tone bursts with a fast onset slope (Pantev et al., 1991). Our intention was studying a ‘small’ response. Therefore we analyzed responses to naturalistic speech sounds, which elicit smaller response amplitudes than brief tone bursts. Estimates of phase statistics from 2400 repeated trials in 42 individuals from dipole source activity in bilateral auditory cortices (total of 84 observations) are shown in Fig. 5 in relation to the simultaneously estimated individual SNRs. An insert in Fig. 5 also shows the time-courses of the grand-averaged gamma responses in left and right auditory cortices with peak amplitudes on the order of 0.5 nAm. The estimates of phase statistics were overlaid by graph of the approximation $n\bar{R}^2 = n \cdot \text{SNR}^2 / (1 + \text{SNR}^2)$. The majority of individual estimates of the phase statistics were located close to, however below the approximating line, which apparently serves as an estimate of an upper limit for the phase statistics.

The individual estimates of the SNR spanned a wide range. The median value was 0.143, the first quartile 0.082 and the third quartile 0.192. The 2.5 and 97.5 percentiles were 0.032 and 0.357, respectively, which is approximately a tenfold difference between smallest and largest SNR. The range of individual values of the $n\bar{R}^2$ phase statistics was between 0.2 and 200. Only for seven out of 84 observations the phase statistics were below the 5% level and

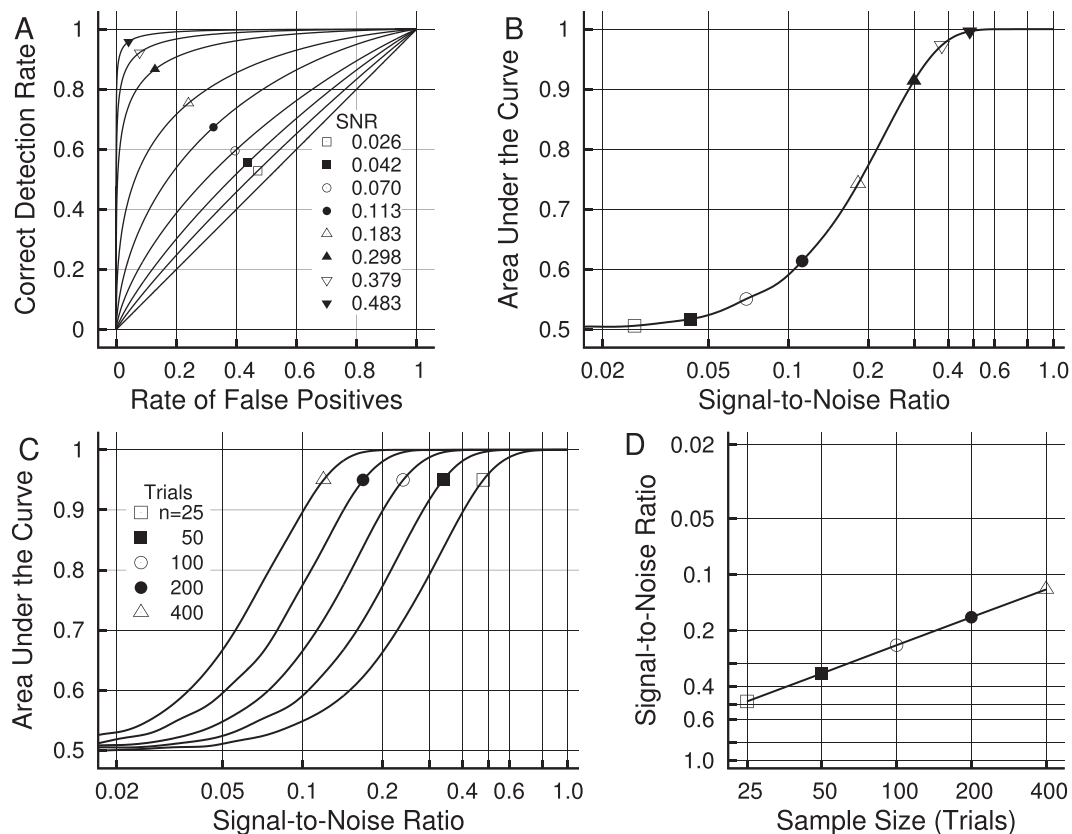


Fig. 6. Performance analysis using the receiver-operating characteristic (ROC). (A) ROC for detecting coherence at various levels of the signal-to-noise ratio (SNR) using the additive data model. With increasing SNR the ROC move from the diagonal, which relates to detection performance at chance level, toward the upper left corner, indicating perfect detection. (B) Areas under the ROC curves providing the probability of detection at a given SNR. The symbols along the graph indicate corresponding ROCs in panel A. (C) AUC characteristics versus the SNR for different sample sizes between $n=25$ repeated trials and $n=400$ trials. The symbols indicate 95% probability of detection. (D) Performance characteristics, showing the SNR, at which detection probability of 95% can be obtained at various sample sizes.

individual signal detection based on the ITC_{PC} was not successful. For 80% of individual observations, the ITC statistics was significant above $p=0.0001$. Calculating ITC_{MSC} using the magnitude-squared coherence resulted in slightly larger phase statistics. The difference was highly significant ($t(83)=12.4$, $p<0.00001$), in only 7 out of 84 cases ITC_{PC} was superior to ITC_{MSC} .

3.5. Performance analysis

Fig. 6 demonstrates the course of performance analysis of synchrony detection and explains how the performance graphs were obtained. Repeated simulations of the coherence measures under H_0 and H_1 resulted in a series of ROCs with the parameter SNR or the phase-mixing factor α (Fig. 6A). Calculating the AUC for each ROC resulted in a characteristic of the total probability of detection versus the SNR (Fig. 6B) and a series of such characteristics according to parametric variation of number of trials or signal duration (Fig. 6C). Selecting the point of 95% probability of detection finally resulted in the performance graph shown in Fig. 6D. The performance graph shows the SNR, which is required at a given sample size for obtaining 95% probability of detection. Thus, the SNR became the dependent variable and spans the y-axis in Fig. 6D rather than the x-axis in Fig. 6B and C.

3.6. Additive signal model

The AUC-characteristics, indicating total probability of detection, showed in general a step-like behaviour (Fig. 7A and B). Specifically the PLV characteristics increased from chance level to 95% likelihood of detection when the SNR increased just by a factor

of two. For the further analyses, we defined a performance indicator by the minimum SNR which is required for 95% probability of detection. The SNR at the 95%-level is shown as function of sample size and signal duration in Fig. 7C and D.

Fig. 7C shows that ITC using either the PC or MSC algorithms required an SNR in the range between 0.025 ($n=800$ trials) and 0.2 ($n=12$ trials) for successful detection. In order to obtain detectability at twofold smaller SNR, four times the number of trials would be required. The performance for the PLV, using PC or MCS algorithms was noticeably lower than for ITC ; PLV required a SNR in the range between 0.18 ($n=800$) and 0.5 ($n=12$). A 16-times larger number of trials would be required in order to obtain detectability at twofold smaller SNR. Performance of MSC was slightly superior to performance of PC, a 7.7% larger SNR was required when using the PC algorithm for ITC ; equivalently, 15% more trials would be required in order to reach the same performance as calculating ITC with the MSC algorithm. For PLV, PC required 5.4% higher SNR or 23% more trials for same performance as the MSC algorithm. The mutual-information method (MI) was less sensitive than PC and MSC; for PLV it required 23.5% larger SNR than MSC for same performance, which is a quite large difference because 2.3 times more trials would be required for the mutual information method. For ITC , the MI method performed substantially poorer than PC and MSC; MI required 1.6 times larger SNR than MSC at $n=12$ trials and even 3.2 larger SNR at $n=800$ trials for equal detection performance.

Performance comparison for the additive model with the signal duration as parameter showed similar relations between the algorithms as discussed for variation of the sample size (Fig. 7D). For obtaining same detection performance at k -times lower SNR, k^2

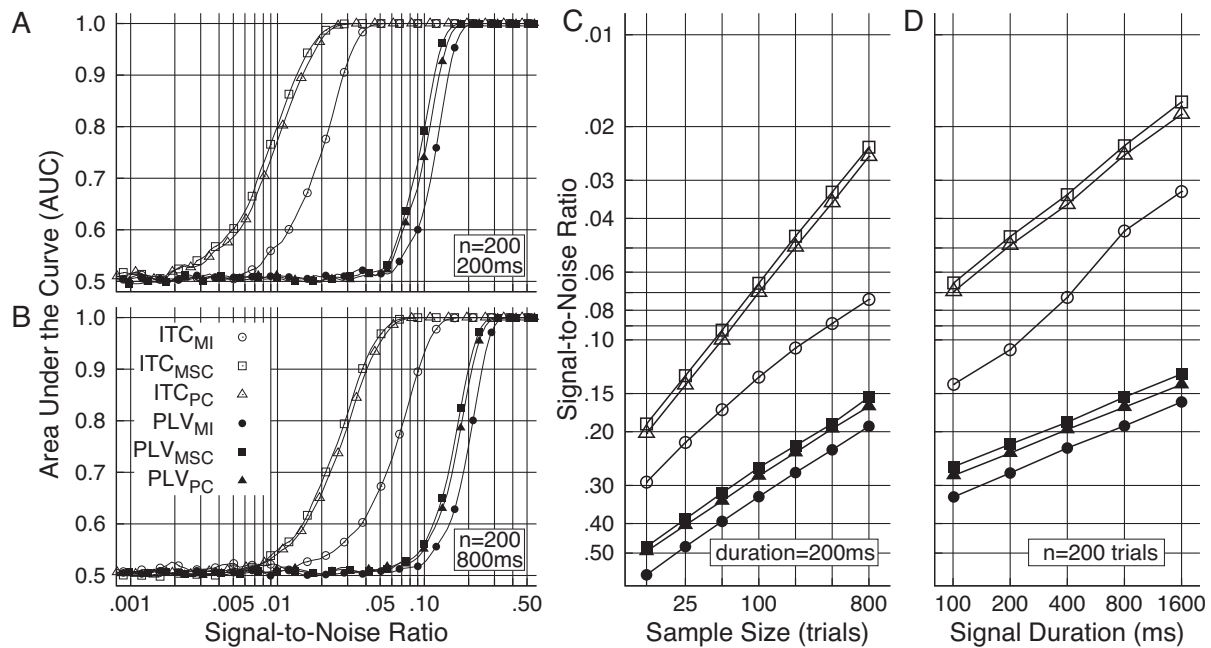


Fig. 7. Performance of detecting synchrony as inter-trial coherence (ITC) or phase-locking value (PLV) using different algorithms of phase coherence (PC), magnitude-squared coherence (MSC) and the mutual information (MI) under the assumption of an additive model of signal generation. (A and B) Characteristics of the area under the ROC curve (AUC) in relation to the signal-to-noise ratio (SNR) for signal duration of 200 ms and 800 ms and $n=200$ repeated trials. (C) Performance characteristics as function of the sample size. In general performance was higher for ITC than PLV and improved with increasing signal duration and number of trials. For the additive signal model, MSC performance was slightly higher than PC, both superior to the MI method. Detection performance is higher if successful detection can be obtained at lower SNR therefore, the y-axis shows a decreasing SNR scale. (D) Performance characteristics as function of the signal duration. The symbols indicate the SNR, at which the AUC characteristics reach the 95% level.

longer signal duration would be required for ITC calculation using PC or MSC, and k^4 longer signal duration for PLV calculation with all methods. Again, the simulation showed that MSC performed slightly better than PC, and a wide gap exists between ITC detection and PLV. It seemed that the mutual information method benefits from longer signal duration when estimating ITC; the lack in performance compared to MSC and PC was smaller for long duration than for short.

The largely different performances for ITC and PLV were demonstrated with time courses of ITC and PLV statistics for the example data of auditory evoked transient gamma responses (Fig. 8). The 42 participants were ranked according to the SNR (Fig. 5), assigned to sub-groups of ten participants, and group averages of ITC and PLV were calculated. Although the ITC values were widely variable, group average ITC was highly significant even for the smallest SNR values (Fig. 8A). Phase-locking between both auditory cortices resulted in generally smaller statistics and significance was reached only in the sub-groups of participants with SNR above the median value.

3.7. Phase-reset model

The AUC characteristics for the phase-reset data model (Fig. 9A and B) again showed step-like behaviour similarly as for the additive model. However the performance of the various algorithms was noticeably different. When comparing the SNR required for 95% probability of detection for different sample sizes (Fig. 9C), the MSC algorithm required an SNR of 0.023 ($n=800$) to 0.20 ($n=12$) for ITC estimation and 0.16 ($n=800$) to 0.50 ($n=12$) for PLV, which are similar ranges as found for the additive model. Using the phase-coherence method showed a significant advantage over MSC. MSC required 1.85 times larger SNR than PC for ITC and a 3.4 times larger sample size would be required for the MSC method than PC for obtaining same performance. Similarly, MSC required 1.81 times

larger SNR than PC for PLV. This difference is more crucial for PLV compared to ITC detection, because the characteristic of SNR versus the sample size is shallower for PLV than ITC, and a 10.8 times larger sample size would be required for the MSC method for obtaining same performance of PLV detection. Also estimating PLV with the MI method was more sensitive for the phase-reset model compared to the additive model. Performance of MI was more similar to PC than MSC.

3.8. Phase-interaction model

The performance characteristics for ITC in case of the phase-interaction were identical with the ITC characteristics observed for the phase-reset model (Fig. 10). This is immediately comprehensible because the ITC case reflects phase-interaction with a source of constant phase; this is how we modeled the phase-reset. A more important result was that the PLV characteristic matched the ITC characteristics for PC and MSC. PLV performance using mutual information method was even superior to the ITC performance. The good detection performance for PLV is in dramatic contrast to the results found for the additive and phase-reset model: Only for the reciprocal phase-interaction, detecting PLV was equal sensitive as signal detection using ITC.

3.9. Performance of phase-lagging index

Performance characteristics for the PLI in comparison to PC, MSC and MI are shown for the additive data model in Fig. 11 and demonstrate as a predominant effect that PLI performance increased monotonously with increasing size of the phase lag to a maximum at $\Delta\varphi = \pi/2$. For small amounts of phase lag, PC, MSC, and MI performances were not largely different and required a SNR larger than 0.2 for successful detection, which is consistent with the characteristics in Fig. 7 for sample size of 200 trials and signal

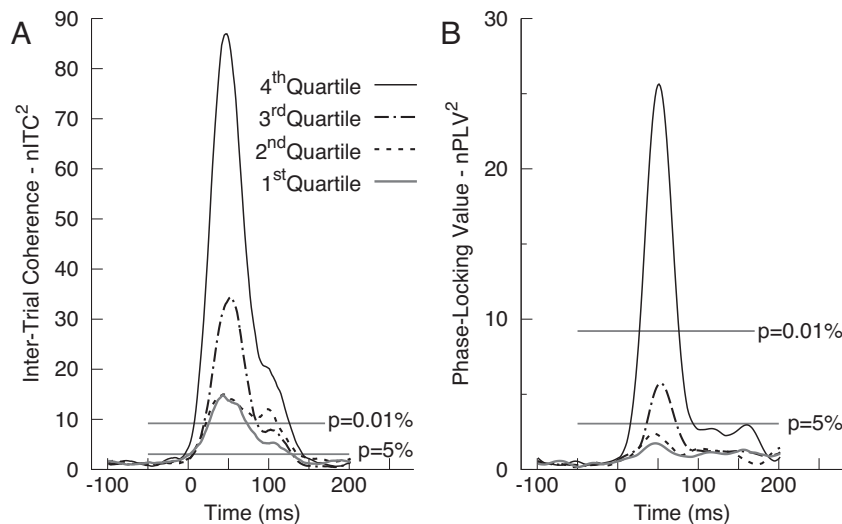


Fig. 8. Time courses of ITC (A) and PLV (B) for the auditory evoked transient gamma band data shown in Fig. 5. Participants were ranked according to their SNR, and group averages were calculated for sub-groups of ten participants (1st to 4th quartile). While signal detection using ITC was successful in all sub-groups, synchrony between left and right auditory cortex measured with PLV was significant only for participants with SNR larger than the group median.

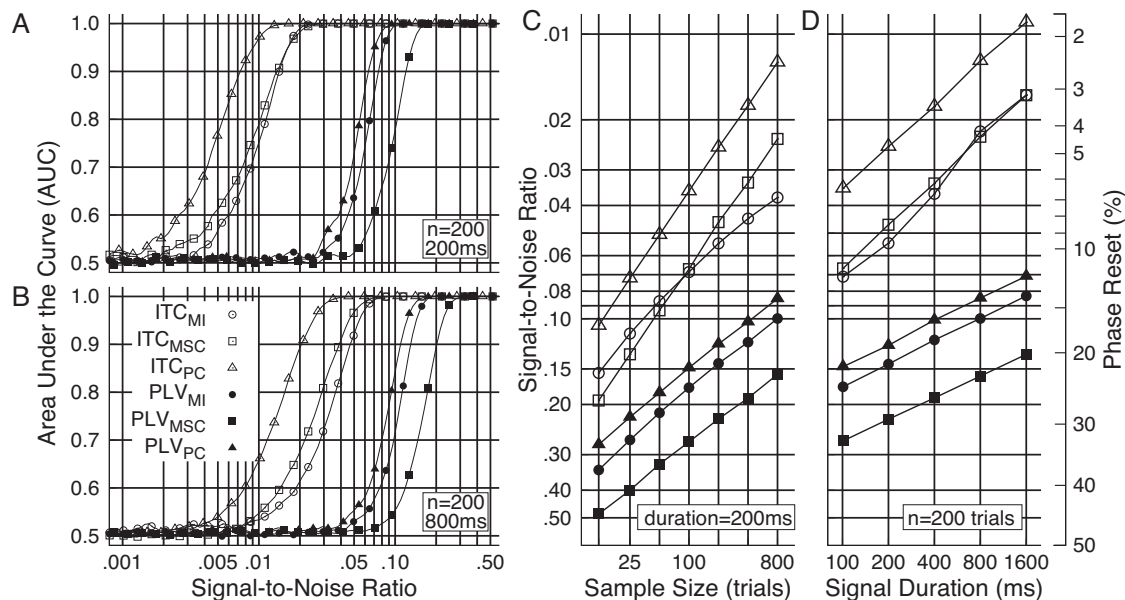


Fig. 9. Performance of detecting synchrony under the assumption of a phase reset mechanism of signal generation. Whereas the performance of MSC based algorithms was similar for the additive and phase-reset signal models, PC and phase-space based algorithms performed noticeably better.

duration of 200 ms. PLI required a three times larger SNR at the smallest phase lag of $\Delta\varphi = 6^\circ$, and only when the phase lag was $\Delta\varphi \geq \pi/4$ PLI performance was superior to the other algorithms. The PLI advantage became especially evident for large correlation between the real parts of the signals y_1 and y_2 .

For the phase-reset data model, in general similar performance characteristics were obtained (Fig. 12). However PC and MI showed a clear advantage over MSC comparable to the characteristics shown in Fig. 9. Most importantly, PLI was more sensitive and required approximately half the SNR compared to the additive data model. Nonetheless, the relation of PC and PLI performance characteristics was similar for phase-reset and additive model, and a clear advantage of PLI was found for large phase lag and underlying correlation in the signals.

For the phase-synchronization model, we found again a substantial increase in performance compared to the phase-reset and additive data model. Therefore, the SNR scale in Fig. 13 had to

be changed to smaller values. In contrast, the sensitivity of PLI improved less from the change in the data model than the other methods. Superior performance of PLI was found for large phase lags of $\Delta\varphi = \pi/2$ only, and the advantage over the other methods was even larger for large correlation between the signals y_1 and y_2 .

Most notably the sensitivity of PLI was not affected by correlation in the simulated signals for all three data models. This makes PLI robust against signal spread effects, which was a main motivation for developing the PLI method.

4. Discussion

Our performance analysis of synchrony detection in EEG and MEG was focused specifically on the role of the underlying models of signal generation. This approach gave important insights into

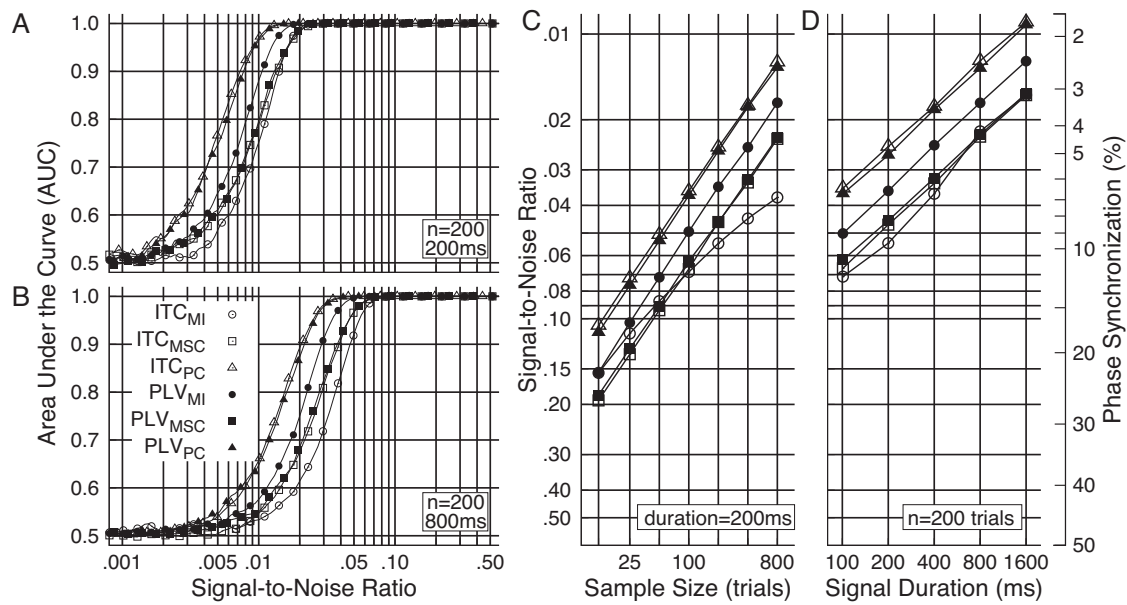


Fig. 10. Performance of detecting synchrony under the assumption of a phase synchronization mechanism of signal generation.

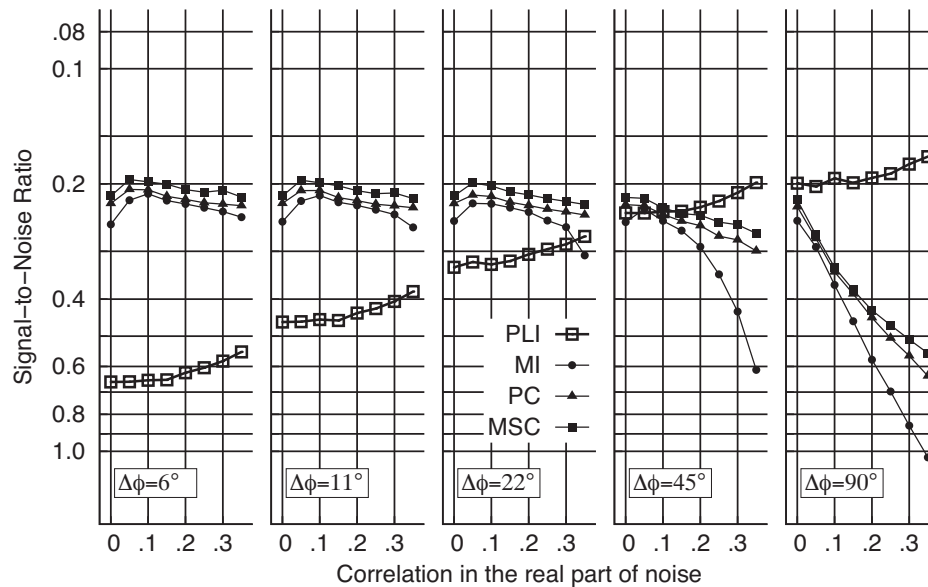


Fig. 11. Performance of detecting synchrony between two signals y_1 and y_2 at various levels of correlation in the real parts of both signals under the assumption of an additive signal model. The phase difference $\Delta\phi$ between y_1 and y_2 was introduced as additional parameter to investigate the performance of synchrony detection based on the phase-lagging index (PLI). Specifically for $\Delta\phi = \pi/2$ the PLI performed superior compared to the other algorithms. Moreover, PLI performance was only weakly affected by correlation between the real part of the noise contribution to y_1 and y_2 while performance degraded strongly for the other algorithms, specifically when the phase lag was larger than $\pi/4$.

possible limitations of synchrony measures and provided support for their interpretation.

Comparison between reports of synchrony measures is often compromised by the fact that the sample mean of phase statistics depends on the sample size, and sometimes the degree of freedom in the data of respective studies is not sufficiently documented. Therefore, we presented as a first step of our analysis a modified test statistics, which is independent of the sample size and provided another advantage that it is universally valid for ITC measures based on the signal phase as well as PLV measures based on the cross-spectral density. The $n\bar{R}^2$ values are comparable to z-scores in linear statistics, and p-values can be estimated easily. Preferably $n\bar{R}^2$ values should be reported as result of phase statistics rather than raw coherence values.

Although the Rayleigh test is most commonly used as parametric test of phase statistics, several alternatives have been developed, namely the Hodges-Ajne test, Kuiper's test, and Watson's U2-test (Mardia, 1972). However, closely similar performance has been reported (Cebulla et al., 1996) with the exception of some advantage of a modified Rayleigh test (Moore, 1980), which uses weighted phase values, a similar concept as has been introduced to the phase-lagging method (Vinck et al., 2011).

When using parametric tests for inferring significance directly from the results of phase statistics, one caveat is, that the assumption of data under the null-hypothesis has to be valid. This may not be the case in any EEG and MEG application because physical signal spread causes some level of dependency in the considered signals. In some studies, surrogate data under the null hypothesis

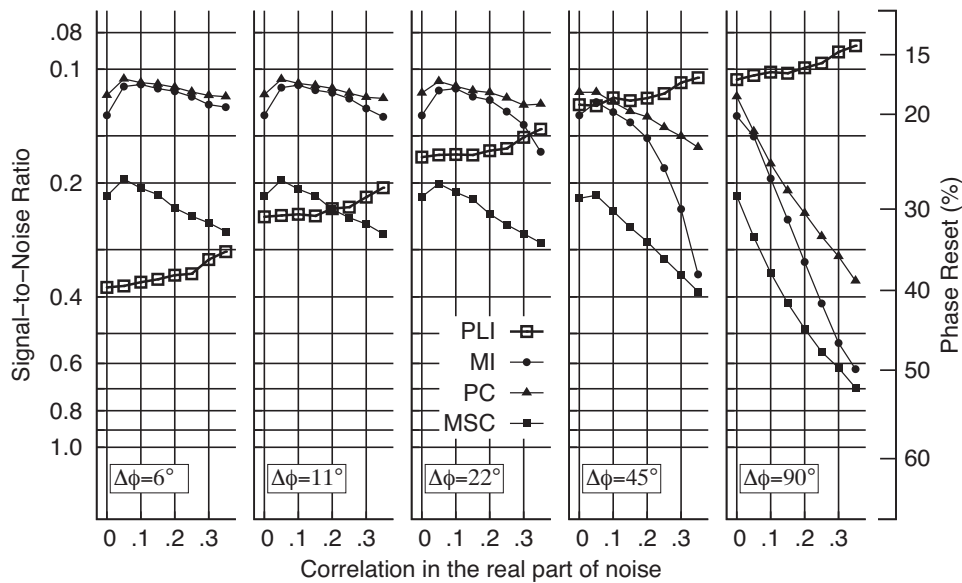


Fig. 12. Performance of detecting synchrony between two signals y_1 and y_2 at various levels of correlation in the real parts of both signals under the assumption of a phase-reset signal model.

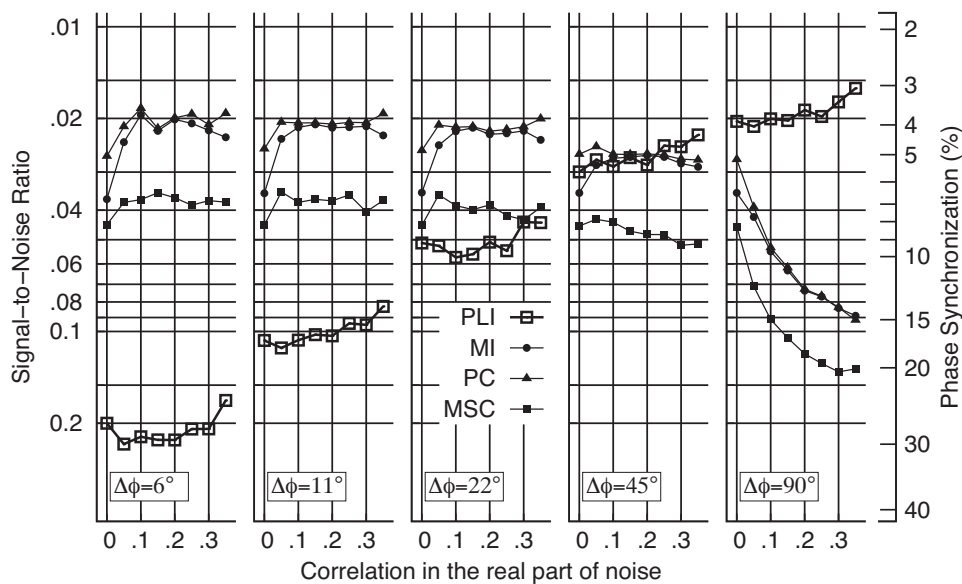


Fig. 13. Performance of detecting synchrony between two signals y_1 and y_2 at various levels of correlation in the real parts of both signals under the assumption of a phase-interaction signal model.

are generated using permutation or phase scrambling (Sun et al., 2012). However this may not solve the problem of a bias introduced by signal spread. Therefore additional statistical tests are required for testing the significance of synchrony differences between experimental conditions or between time intervals related to the experimental procedure, preferably those are non-parametric tests (Maris et al., 2007).

We used the ROC method for our performance analysis, which compares an outcome measure under H_1 with the null-hypothesis H_0 . Again, if the assumptions of a clear contrast between null hypothesis and the alternative are not entirely fulfilled in the experimental data, the ROC method informs about performance differences between methods but does not explain how a certain performance advantage can be realized.

The SNR was a major factor affecting synchrony detection. Performance analyses for all types of signal models and all algorithms resulted in steep, step-like characteristics of total probability of detection versus the SNR. Commonly, an SNR threshold exists, which makes detection almost certain for SNR values above threshold and makes synchrony detection almost impossible below. In contrast to the small interval in SNR, which defines the transition between success and failure of detection, we found a wide range of SNR in an example set of MEG data. Although ITC detection was possible in more than 90% of individual cases, an analysis method requiring a larger SNR will fail in an unfortunate large number of individuals. This seems a severe issue especially because we found that PLV detection likely requires a larger SNR than ITC detection. The characteristic behaviour of synchrony measure, that they may

be successful in some individuals but fail in others, calls for new concepts of group analysis, rather than reporting effects as the group mean.

We compared ITC versus PLV measures, which may be seen as an unusual comparison of different entities like comparing apples and oranges. Our intention was to use ITC, which has been proven as reliable measure of signal detection, to get information about the minimum required SNR. Then our question was, whether signal detectability would be a sufficient prerequisite for PLV, or whether PLV detection would even be possible at lower SNR. It turned out that PLV detection required at least the same level of SNR as ITC; more likely PLV is less sensitive. For the additive and phase-reset models we found that substantially larger SNR was required for PLV than for ITC. The large performance difference between ITC and PLV may not have been presumed easily. For our example data of auditory evoked 40-Hz responses it means, that we detected the low amplitude 40-Hz oscillations, which appear as clearly synchronous oscillations in bilateral auditory cortices in the averaged source signal (Figs. 5 and 8). Bilateral auditory cortices were innervated by a common input. Even if both auditory cortices did not interact directly with each other, the common input fulfills the criteria for the additive signal model. However, while signal detection was possible for almost all individuals, detection of synchrony between left and right auditory cortex was more difficult and failed in most individuals.

Performance was superior for MSC compared to PC in case of the additive signal model, whereas the reverse was found even with stronger contrast between methods for the phase-reset model. These results seem to be reasonable; considering amplitude information when using the MSC algorithm is more appropriate for the additive model and using entirely phase information in the PC algorithm fits better to the phase-reset model than vice versa. The question would be whether such properties could help to conclude about the underlying signal model.

Performance advantage of MSC over PC has been previously reported for the detection of auditory brainstem responses and stimulus induced 40-Hz responses (Cebulla et al., 1996; Dobie and Wilson, 1994) however the MSC advantage was small and sometimes even not noticeable. Detecting ITC in our example data of auditory evoked gamma responses showed also small but significantly larger test statistics for MSC compared to PC, which corroborates the earlier reports. The advantage of MSC over PC is consistent with our performance analysis for the additive signal model. Again, would such performance contrast be sufficient to conclude an additive signal model? Although MSC results were superior to PC, Dobie and Wilson (1994) speculated even in favor of a phase reorganization model because the MSC advantage over PC was smaller than expected. The authors compared their experimental data with simulations, using an additive signal model, and showed similar characteristics of the area under the ROC versus the SNR, like in our study. However, they did not conduct simulations with a phase reorganization data model, for which we found an advantage of PC over MSC. Noticeably the PC advantage for the phase reorganization model was substantially larger than the contrast between methods for the additive model. Thus, it might be possible, interpreting clearly superior performance for the PC method as supporting a phase-reset data model. Interestingly, reports about comparison between MSC and PC show commonly a marginal advantage of MSC (Simpson et al., 2000). It might be that most of such comparisons have been conducted for detection of primary sensory responses, for which the additive data model may be valid, although most authors prefer to assume a phase-reorganization model. It would be interesting to see future comparisons between methods for a wider range of signal types in EEG and MEG.

Nonetheless, our analysis demonstrated that phase statistics using entirely phase information are also sensitive in case of a purely additive signal model. While it might be possible to infer the underlying data generation model from different behaviour of synchrony measures, it would be a severe fallacy, concluding phase reorganization simply because phase statistics had been used.

The effect of noise on the performance of synchrony measures has been studied before with surrogate and real EEG and MEG data. When comparing synchrony measures for recordings from the cortical surface in left and right hemispheres in rats, equal performances were found for several linear measures using circular variance, similarly as in our study, however performance was lower for mutual information based approaches, for which the authors assumed better performance for application to very long sequences of data (Quiroga et al., 2002). Those results are consistent with our simulations using the additive model, which showed inferior performance for a mutual-information algorithm with a tendency to better performance for signals with longer duration.

Methods of non-linear dynamics have been introduced with great enthusiasm to EEG and MEG signals analysis under the assumption that the dynamics of information exchange between neural populations might be strongly non-linear rather than being based on linear systems. Related performance analyses have been conducted mainly with physical models such as coupled non-linear oscillators (Kreuz et al., 2007) or a Rössler system with superimposed white noise (Xu et al., 2006). Again, an open question is whether comparison between linear and non-linear methods could inform about the underlying data generation mechanism. Commonly the studies on detection of non-linear phenomena did not consider detection performance on linear data models.

The suggestion, that volume conduction in EEG affects the real part of coherence but not its imaginary part (Nolte et al., 2004), inspired new approaches to analyses of synchrony and led to the development of PLI (Stam et al., 2007). Several reservations have to be considered when discussing possible benefits from the PLI method. First, even if real and imaginary parts of coherence are widely independent, as reflecting mainly volume conduction in the real part and true neural interaction in the imaginary part, the phase is not independent of both factors: Given a constant imaginary part, the phase will become smaller, if the real part increases. Assuming that the effect of volume conduction may be dominant, the phase is small by definition. Second, brain connectivity occurs as reciprocal interaction and is not necessarily accompanied by a time delay. Thus, phase differences between signals may be small, and our analysis showed poor performance in case of a small phase lag, compared to other data analysis approaches. Regardless, whether PLI may allow of separating volume conduction and true neural communication, discarding the real part and concentrating on the imaginary part instead or its sign, constitutes a loss of information. This again could be interpreted as reduction in SNR and consequently lower performance. Low sensitivity of PLI in case of noise and small phase lags led to development of a phase-weighted version of the PLI (Vinck et al., 2011), which preferably included data with phase lag of $\pm\pi$, which is consistent with our finding, that the performance of PLI was optimal for a phase lag of $\pm\pi$. But then again, the question is how realistic a phase lag in this range would be, specifically in noisy data. In case of a phase lag close to $\pm\pi$, the real part of synchrony is very small compared to the imaginary part, which contradicts the initial assumption that signal spread would contribute to the real part, which in terms could become substantially larger than the imaginary part. Thus, in general we should assume a relatively small phase lag, for which we found relatively poor performance of the PLI method. However, our analysis focused on thresholds of detectability in noise; advantages of PLI over other methods may come into play at higher SNR.

We found superior performance for coherence detection in noise when true phase interaction was the underlying neural mechanism. The advantage over the phase-reset and additive signal models was tremendous: It seems that in case of low SNR, coherence can be detected for true phase interaction only and seems impossible in case of the other signal models, even with endless number of repeated trials. This should not be a severe problem, because reciprocal phase interaction and synchronization of local oscillators have been suggested recently. Indeed, the role of the phase of such oscillatory activity for neural timing and consequently for effective functional connectivity is currently a hot topic in neurosciences. However, it might be sometimes overlooked, that the basic mechanism of phase-reorganization and neural timing are grounded on experimental findings from single neuron recordings and models of small neural populations. However, the neural mechanism of phase organization at microscopic level may not be visible easily at the macroscopic level of mass activity in EEG and MEG. It might even be that phase organization at microscopic level appears as not distinguishable from an additive model at the macroscopic level, as it has been suggested from a recent simulation study (Telenczuk et al., 2010). The classical example for this point of view is the concept that changes in synchrony within a neural population results in externally measurable signal power changes in EEG and MEG, which led to the term ‘event-related synchronization’ (Pfurtscheller and Lopes Da Silva, 1999).

Finally, how to obtain a high SNR? Obviously, the answer depends on the frequency of interest. Because of the $1/f$ -characteristics of the EEG and MEG spectrum, low frequency signals are largest and show most likely high SNR. For high frequency oscillations, one may capitalize on entrainment methods, which elicit oscillations at the frequency of sensory stimulation and can be obtained with high SNR. Analysing the effect of the SNR on coupling between low frequency phase and high frequency amplitude could become complicated because of different SNR for the signal components at different frequencies. Related performance analyses of cross-frequency coupling with respect to SNR have still to be conducted.

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